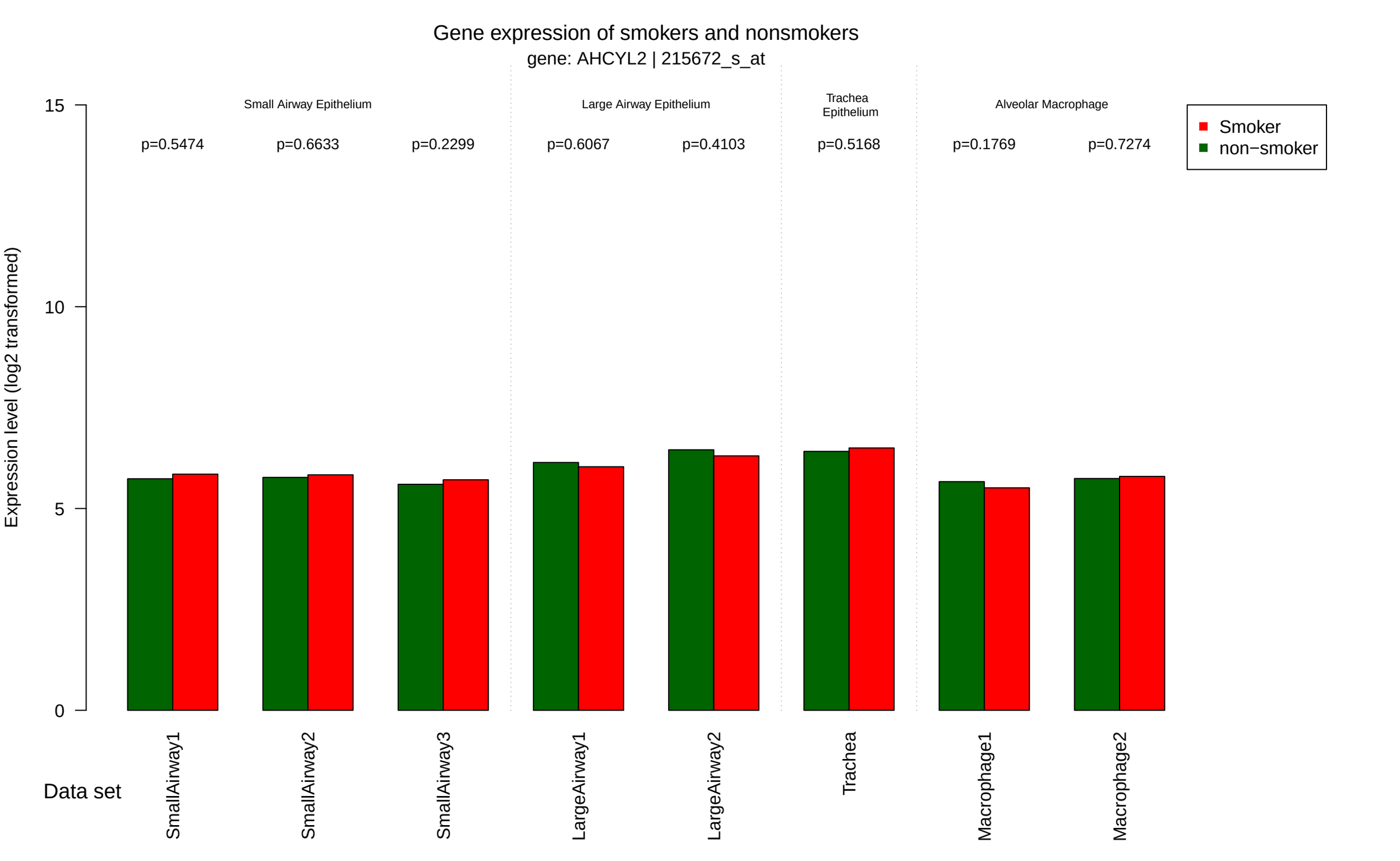
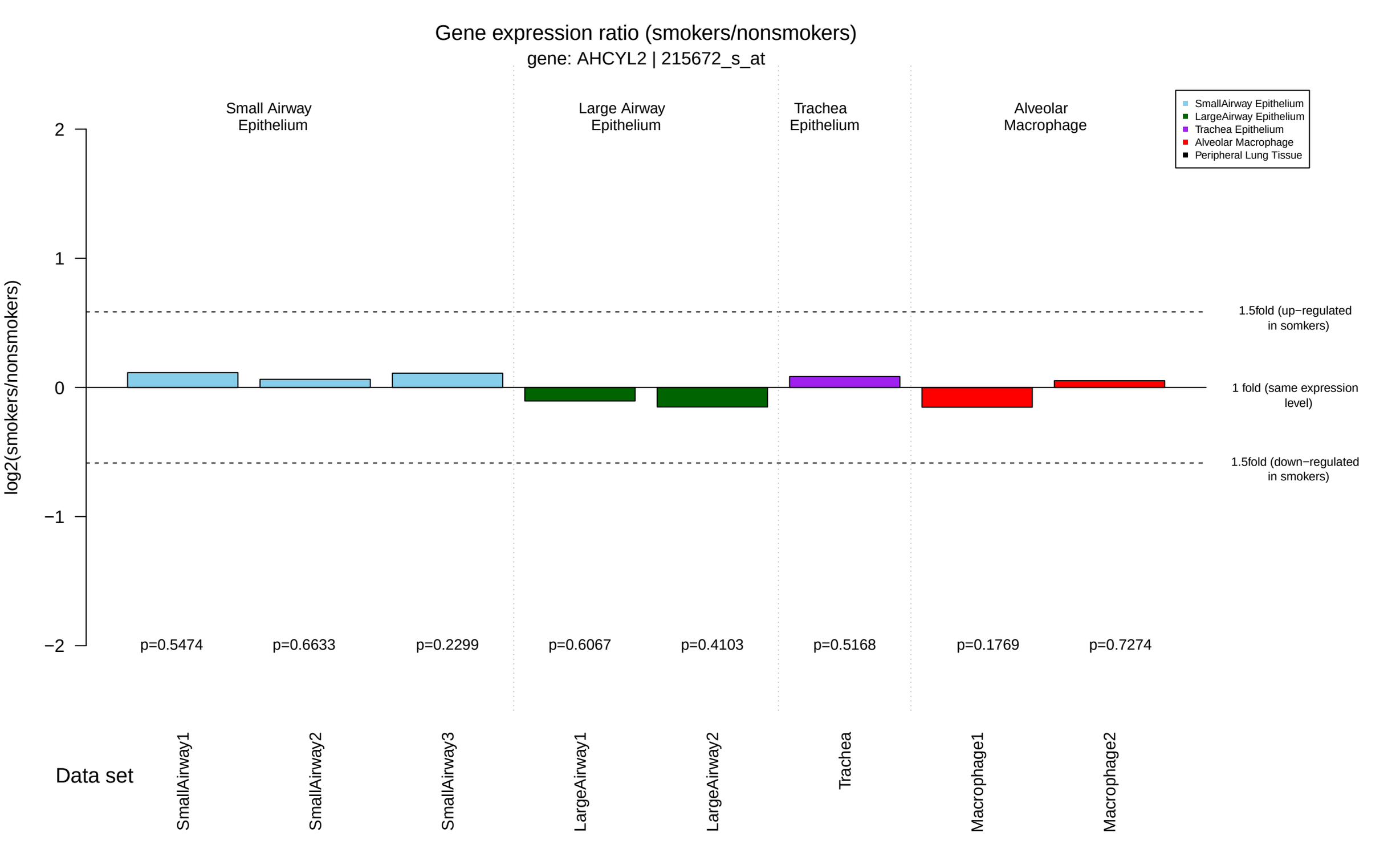
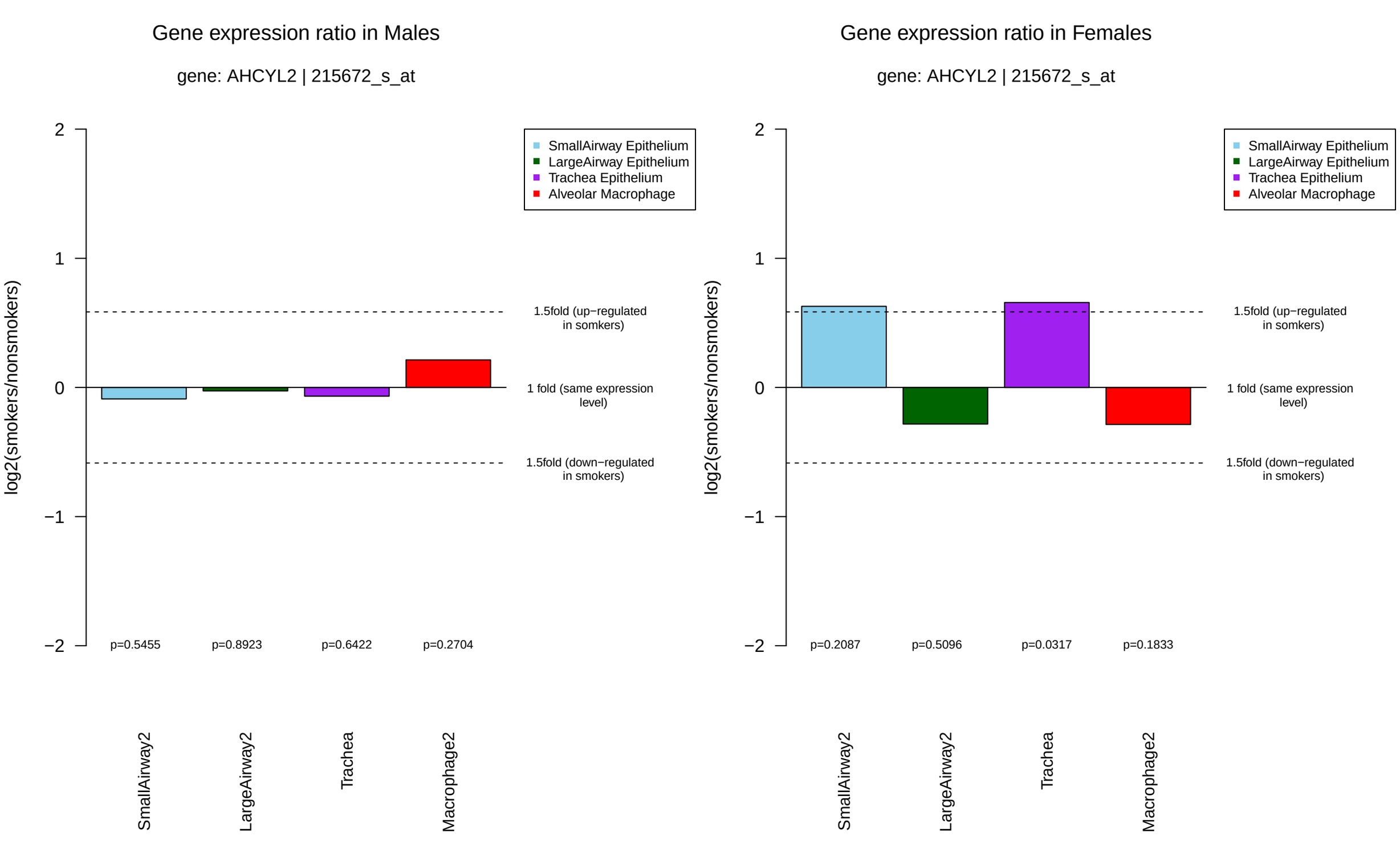
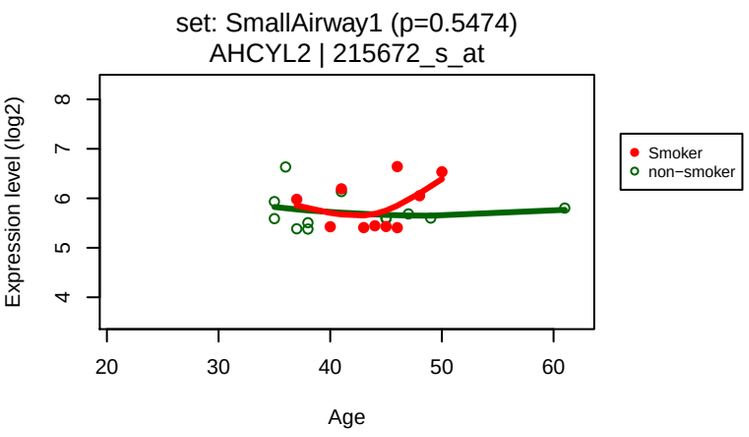
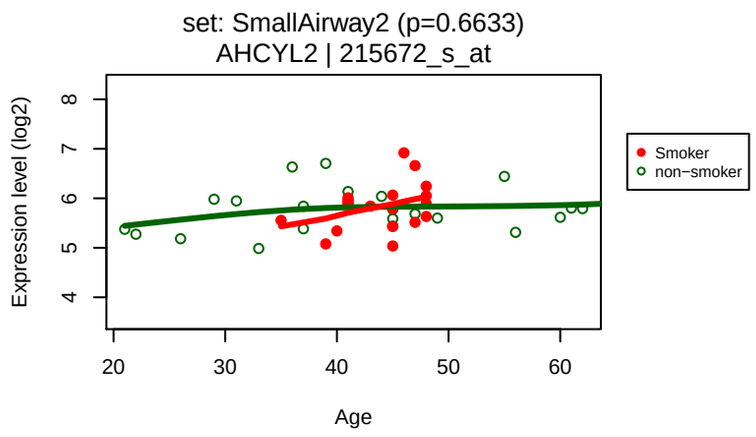
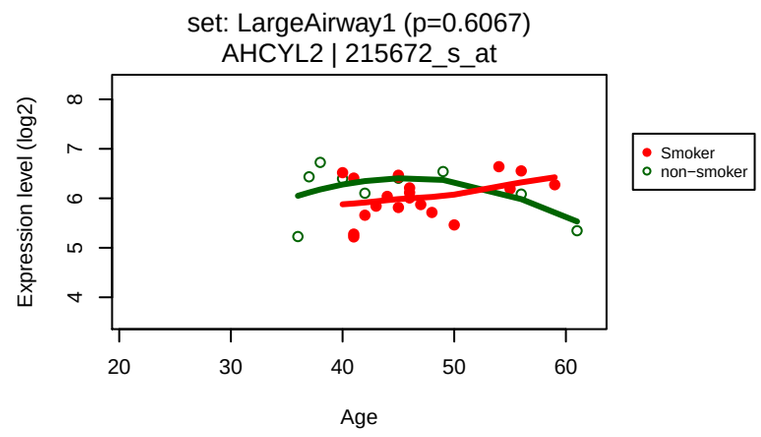

Gene expression of smokers and nonsmokers
gene: AHCYL2 | 215672_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5474
p=0.6633
p=0.2299
p=0.6067
p=0.4103
p=0.5168
p=0.1769
p=0.7274
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

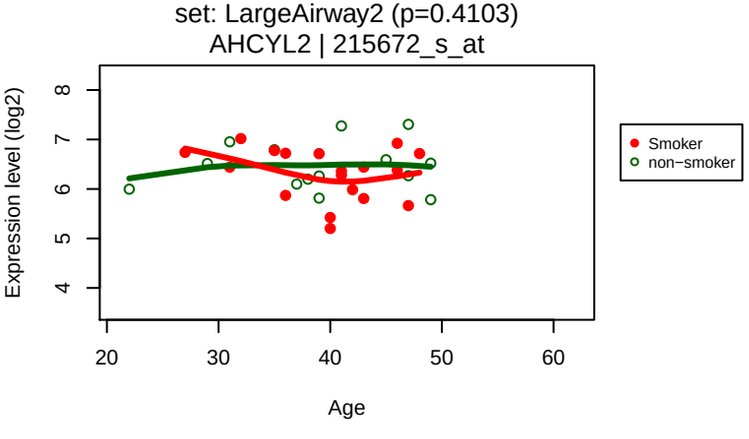

Gene expression ratio (smokers/nonsmokers)
gene: AHCYL2 | 215672_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5474
p=0.6633
p=0.2299
p=0.6067
p=0.4103
p=0.5168
p=0.1769
p=0.7274
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

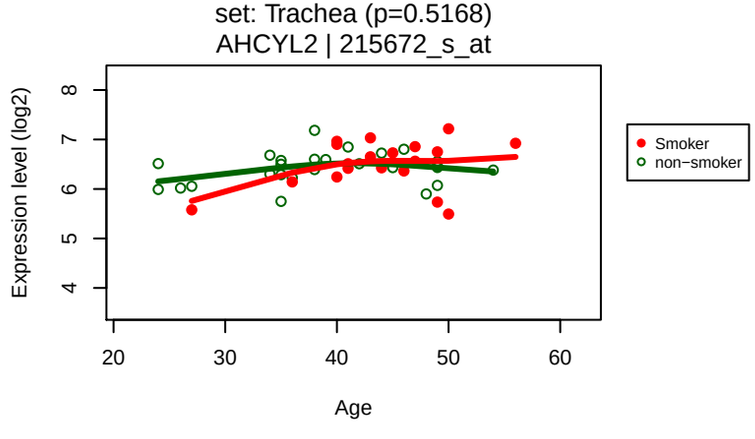

Gene expression ratio in Males
gene: AHCYL2 | 215672_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5455
p=0.8923
p=0.6422
p=0.2704
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AHCYL2 | 215672_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2087
p=0.5096
p=0.0317
p=0.1833
SmallAirway2
LargeAirway2
Trachea
Macrophage2

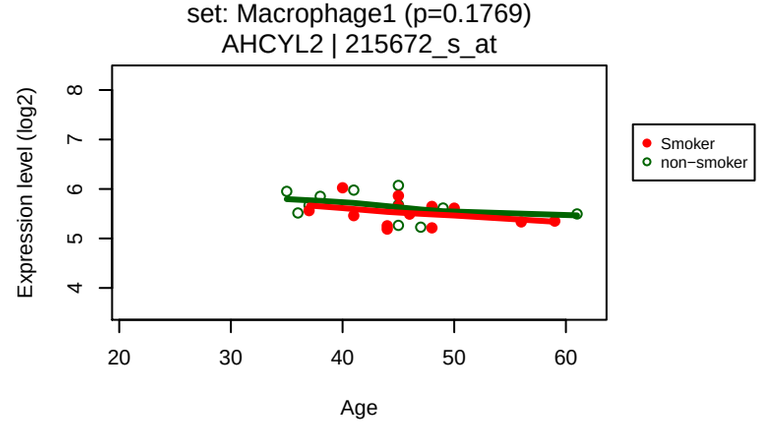

set: SmallAirway1 (p=0.5474)
AHCYL2 | 215672_s_at
Expression level (log2)
Age
Smoker
non−smoker

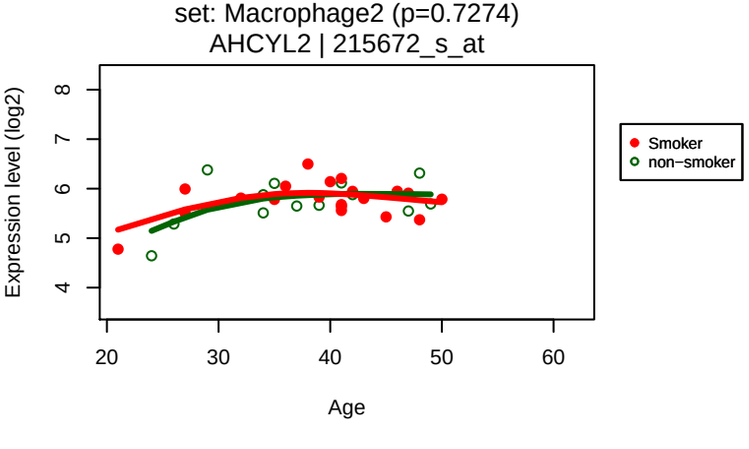

set: SmallAirway2 (p=0.6633)
AHCYL2 | 215672_s_at
Expression level (log2)
Age
Smoker
non−smoker

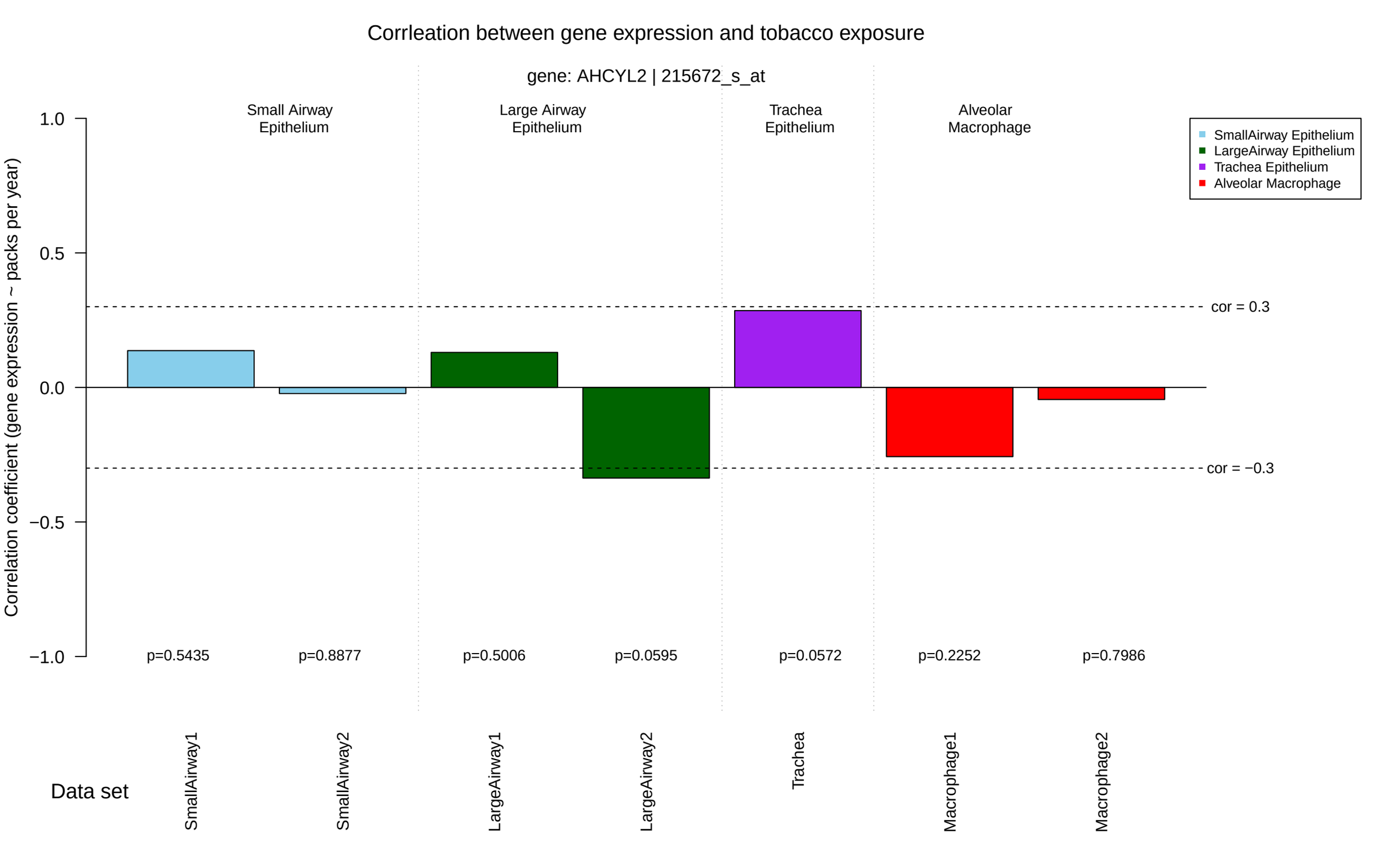

set: LargeAirway1 (p=0.6067)
AHCYL2 | 215672_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4103)
AHCYL2 | 215672_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5168)
AHCYL2 | 215672_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1769)
AHCYL2 | 215672_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7274)
AHCYL2 | 215672_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AHCYL2 | 215672_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5435
p=0.8877
p=0.5006
p=0.0595
p=0.0572
p=0.2252
p=0.7986
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2